

# Faculty Guide Selection System

Hackathon Project Brief • II MBA • Department of Management Studies

[faculty-selection-bitsathy.vercel.app](https://faculty-selection-bitsathy.vercel.app) ↗

**Project Vision:** A high-speed, fair, and tamper-proof web portal for MBA students to select their project guides on a strict **First-Come, First-Served (FCFS)** basis. Eliminates manual spreadsheet collisions, double-booking, and seat disputes with automated real-time tracking.

## ⚡ THE TECHNOLOGY STACK (SIMPLE & RELIABLE)



### Modern Web App

Ultra-light Single Page App hosted on Vercel Edge. Sub-300ms response time on 4G/5G mobile.



### Google OAuth

1-tap login restricted to college accounts (@bitsathy.ac.in). Zero fake logins or spoofing.



### Supabase Backend

Cloud PostgreSQL database with row locks. Guarantees seats never overflow during traffic rush.



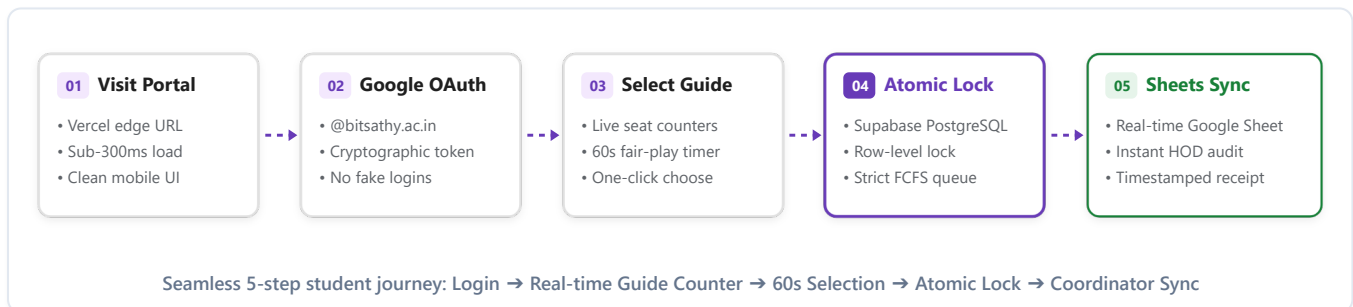
### Google Sheets Sync

Automatic live sync so faculty coordinators view selections in their spreadsheet.

## 📊 CODEBASE ARCHITECTURE & ACTIVE LINES OF CODE

| PART / LAYER                 | TECHNOLOGY         | LINES OF CODE | SHARE | CORE RESPONSIBILITY                                                               |
|------------------------------|--------------------|---------------|-------|-----------------------------------------------------------------------------------|
| Python App Code (Backend)    | Python 3.14        | 1,236         | 54.9% | FCFS row-level lock allocation, Google OAuth JWT verification, Google Sheets sync |
| Frontend Interface (Web App) | Vanilla JS + CSS   | 1,017         | 45.1% | Zero-dependency mobile-first SPA, live capacity meters, tactile Google cards      |
| Total Active Codebase        | Full Stack Runtime | 2,253         | 100%  | Production application codebase deployed on Vercel                                |

## 🔄 END-TO-END WORKFLOW FLOWCHART



## 🔑 KEY PROBLEMS SOLVED

### Zero Seat Overfilling

Database row locks guarantee that the last remaining seat goes to exactly one student.

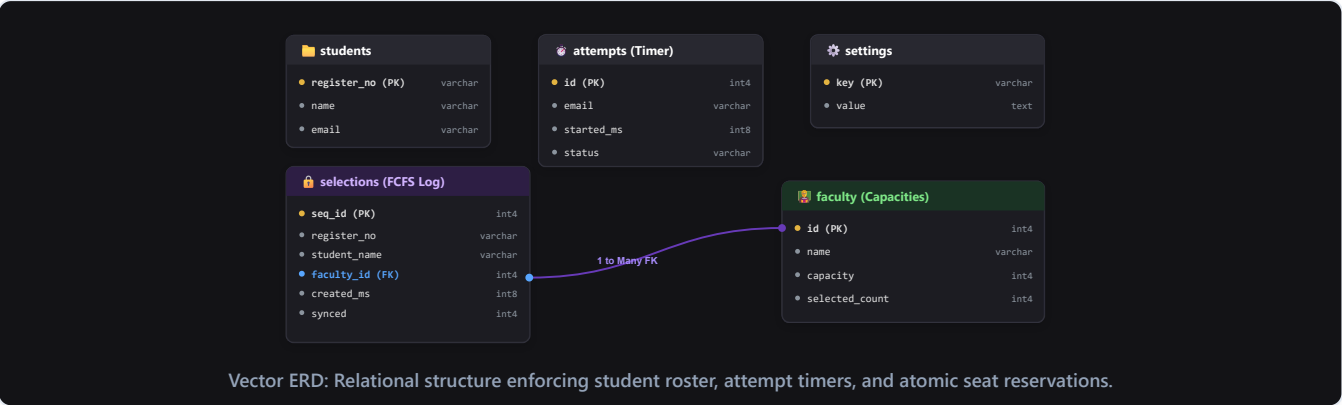
### Fair 60s Timer

Countdown clock prevents students from opening the form and squatting on guides indefinitely.

### No Impersonation

Cryptographic Google token binds each submission directly to the student's verified college identity.

DATABASE SCHEMA (SUPABASE POSTGRESQL)



LIVE PORTAL INTERFACES & AUTOMATED AUDIT

Student Confirmation Screen

Verified

Faculty Selection - II MBA

Your response has been recorded.

suganesans.mb25@bitsathy.ac.in

Switch account

Selection confirmed

|                 |                         |
|-----------------|-------------------------|
| Faculty         | Prof. Suganesh S        |
| Name            | Suganesan S             |
| Register number | 7376257MB144            |
| Submitted at    | 2026-10-06 18:26:16 IST |
| Response no.    | #1                      |

Selections can't be changed. Contact your coordinator if something is wrong.

Live Faculty Capacity Counter

Auto-Updated

| Faculty               | # Capacity | # Selected | # Remaining |
|-----------------------|------------|------------|-------------|
| Dr Murugappan S       | 5          | 0          | 5           |
| Dr Adhinarayanan B    | 5          | 4          | 1           |
| Dr Satheesh Kumar T   | 5          | 4          | 1           |
| Prof. Senthil Kumar N | 5          | 5          | 0           |
| Prof. Nandhini B      | 4          | 4          | 0           |
| Prof. Mageswaran J    | 4          | 4          | 0           |
| Prof. Dhanabalu S N   | 4          | 4          | 0           |
| Prof. Saranya S       | 4          | 4          | 0           |
| Prof. Aishwariya M R  | 4          | 4          | 0           |
| Prof. Suganesh S      | 4          | 4          | 0           |
| Total                 | 44         | 37         | 7           |

REAL-TIME GOOGLE SHEETS MASTER AUDIT REPORT (LIVE SYNC)

Official Coordinator Spreadsheet Sync (from II MBA - Google Sheets.pdf)Real-Time Auto-Generated

| # | Seq No | Timestamp (IST)     | Name              | Register No  | Email                               | Faculty              |
|---|--------|---------------------|-------------------|--------------|-------------------------------------|----------------------|
|   | 1      | 2026-10-06 18:26:16 | Suganesan S       | 7376257MB144 | suganesans.mb25@bitsathy.ac.in      | Prof. Suganesh S     |
|   | 2      | 2026-10-06 18:39:34 | Reashma S         | 7376257MB137 | reashmas.mb25@bitsathy.ac.in        | Prof. Suganesh S     |
|   | 3      | 2026-10-06 19:03:13 | Arunachalam S     | 7376257MB105 | arunachalams.mb25@bitsathy.ac.in    | Prof. Aishwariya M R |
|   | 4      | 2026-10-06 19:03:21 | Mahalakshmi T     | 7376257MB125 | mahalakshmit.mb25@bitsathy.ac.in    | Prof. Aishwariya M R |
|   | 5      | 2026-10-06 19:04:07 | Elamathi S        | 7376257MB110 | elamathis.mb25@bitsathy.ac.in       | Prof. Dhanabalu S N  |
|   | 6      | 2026-10-06 19:05:15 | Abishek G         | 7376257MB102 | abishekg.mb25@bitsathy.ac.in        | Prof. Suganesh S     |
|   | 7      | 2026-10-06 19:06:34 | Mathielavarasan S | 7376257MB128 | mathielavarasans.mb25@bitsathy.a... | Prof. Mageswaran J   |
|   | 8      | 2026-10-06 19:06:40 | Yoshidha S        | 7376257MB147 | yoshidhas.mb25@bitsathy.ac.in       | Prof. Mageswaran J   |
|   | 9      | 2026-10-06 19:07:13 | Aswinkumar K      | 7376257MB107 | aswinkumark.mb25@bitsathy.ac.in     | Prof. Suganesh S     |